

Hospital General Information Dashboard - Technical Document

1. Dashboard Pages (Navigation Order)

#	Page Name	Purpose
1	Overview	High-level KPIs, slicers, and summary cards
2	Graphics	Visual charts and graphical breakdowns
3	Performance Measure	Domain-specific performance analysis (Mortality, Safety, Readmission)
4	Scenario Insights	What-If scenario analysis with interactive sliders
5	State and Hospital Details	Drill-down view by state and individual hospital

2. Entity-Relationship Diagram (ERD) & Data Model

The data model uses a **Star Schema** with the **Hospital** table at the center, connected to three dimension tables and one detail fact table.

Tables & Columns

Hospital (Central Fact Table)

Column	Description
FacilityID	Primary Key — unique identifier for each hospital
FacilityName	Name of the hospital
Address	Street address
City	City where the hospital is located
County	County/Parish
BirthingFriendly	Whether the hospital meets birthing-friendly criteria (Yes/No)
EmergencyServices	Whether emergency services are available (Yes/No)

State (Dimension Table)

Column	Description
StateID	Primary Key
State	Two-letter state abbreviation (e.g., TX, CA)
State Full Name	Full state name (e.g., Texas, California)

HospitalType (Dimension Table)

Column	Description
HospitalTypeID	Primary Key
Hospital Type	Display label for the hospital type
HospitalType	Internal type code

HospitalOwnership (Dimension Table)

Column	Description
HospitalOwnershipID	Primary Key
HospitalOwnership	Internal ownership code
Ownership	Display label for ownership type

PerformanceDomain (Detail Fact Table — Unpivoted)

Column	Description
FacilityID	Foreign Key → Hospital[FacilityID]
Domain	Performance category (Mort, Safety, Readm)
Metric	Measure type (Better, Worse, NoDifferent, FacilityMeasureCount)
Value	Numeric score (SUM aggregation)

Relationships

From (One Side)	To (Many Side)	Key
State (1)	Hospital (*)	StateID
HospitalType (1)	Hospital (*)	HospitalTypeID
HospitalOwnership (1)	Hospital (*)	HospitalOwnershipID
Hospital (1)	PerformanceDomain (*)	FacilityID

All relationships use **single-direction cross-filtering** (from the "One" side to the "Many" side).

3. Power Query Steps (Data Transformation)

The following steps were implemented in Power Query to ingest and clean the data:

1. **Connect & Import:** Connected to the source CSV file and imported all columns accurately.
2. **Standardization:** Handled missing or non-numerical values (e.g., replacing "Not Available" with nulls) in the "Hospital overall rating" and measure-count columns.
3. **Data Parsing:** Standardized categorical fields (Hospital Type, Hospital Ownership, Emergency Services) into consistent text labels.
4. **Dimension Table Creation:** Extracted unique values for State, HospitalType, and HospitalOwnership into separate lookup tables with auto-generated surrogate keys (StateID, HospitalTypeID, HospitalOwnershipID).
5. **Validation:** Cleaned and validated ZIP codes, state abbreviations, and the "Meets criteria for birthing friendly designation" field.
6. **Unpivoting (Data Modeling):** Unpivoted the performance measure columns (MORT, Safety, READM, Pt Exp, TE) into a separate PerformanceDomain table with Domain, Metric, and Value columns to facilitate dynamic DAX calculations.

4. List of Key DAX Formulas

KPI Measures:

```
% High-Performing = DIVIDE([Hospitals Above Threshold], [Total Hospitals], 0)
Total Hospitals = COUNTROWS(Hospital)
Avg Overall Rating = AVERAGE(Hospital[OverallRating])
Birthing Friendly % = DIVIDE([Birthing Friendly Hospitals], [Total Hospitals], 0)
```

What-If Scenario Measures:

```
Hospitals Above Threshold = CALCULATE(DISTINCTCOUNT(Hospital[FacilityID]),
FILTER(Hospital, Hospital[OverallRating] >= [Rating Threshold Value]))
Projected Readmit Better = [Readm Better] * (1 + [Readmission Improvement Value])
Projected Safety Better = [Safety Better] * (1 + [Safety Improvement Value])
```

Domain-Specific Measures (Examples):

```
Mort Better % = DIVIDE(
    CALCULATE(SUM(PerformanceDomain[Value]), PerformanceDomain[Domain] = "Mort",
PerformanceDomain[Metric] = "Better"),
    CALCULATE(SUM(PerformanceDomain[Value]), PerformanceDomain[Domain] = "Mort",
PerformanceDomain[Metric] = "FacilityMeasureCount"),
    0
)

Overall Better % = DIVIDE([Mort Better %] + [Safety Better %] + [Readm Better %],
3)
```

5. Instructions for Adding New Performance Domains

If a new performance domain (e.g., "Infection Control") needs to be added to the dashboard, follow these steps:

7. **Update Source Data:** Ensure the new domain columns are included in the source CSV file following the same naming convention (e.g., Infection_Better, Infection_Worse, Infection_FacilityMeasureCount).
8. **Update Power Query:** Open Power Query Editor, go to the unpivoting step for the PerformanceDomain table, and ensure the new columns are selected to be unpivoted into the Domain and Metric columns.
9. **Add DAX Measures:** Create new DAX measures for the domain following the existing pattern:
 - Infection Better % = DIVIDE(CALCULATE(SUM(PerformanceDomain[Value]), PerformanceDomain[Domain]="Infection", PerformanceDomain[Metric]="Better"), CALCULATE(SUM(PerformanceDomain[Value]), PerformanceDomain[Domain]="Infection", PerformanceDomain[Metric]="FacilityMeasureCount"), 0)
10. **Update Aggregations:** Modify the Overall Better % and Best Performing Domain measures to include the new domain in their calculations.